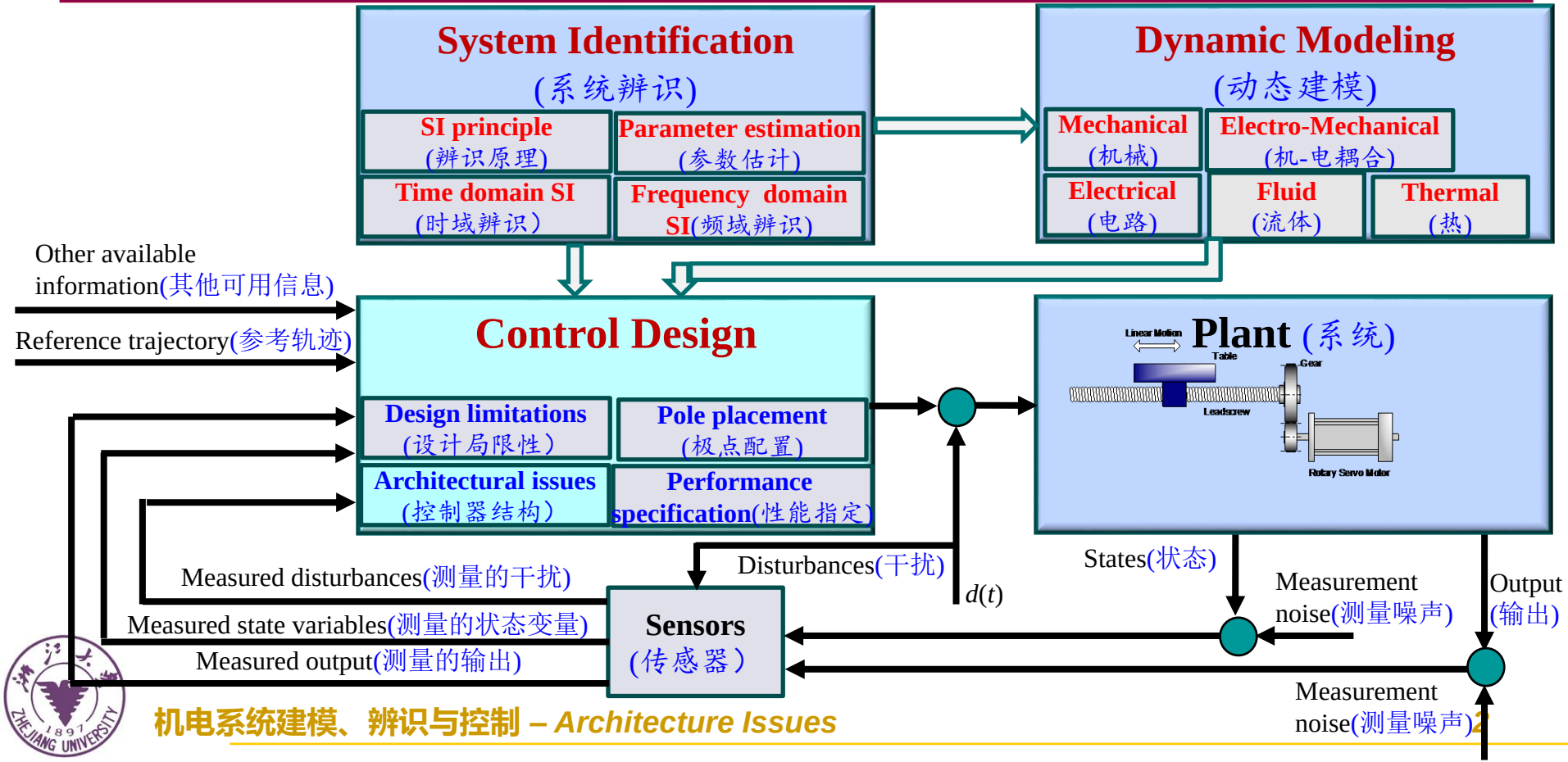
# 机电系统建模、辨识与控制
# Modeling, Identification and Control of Mechatronic Systems

## MIC_L8 Architecture Issues

朱笑丛（Xiaocong Zhu） 副教授

# Main Contents of Course(课程内容)



# Models of Disturbances and References

- **Step Type Disturbances and References**(阶跃类型的干扰和参考输入)

$$d_i(t) = A_{d0}$$

$$\Rightarrow \quad \dot{d}_i(t) = 0 \quad \text{or} \quad P_{dis}(s) = \frac{1}{s} \text{ with } u = 0$$

$$sD_{dis}(s) - d_i(0) = 0 \Rightarrow D_{dis}(s) = \frac{d_i(0)}{s} = d_i(0)P_{dis}(s)$$

- **Sinusoidal Type Disturbances and References** (正弦类型的干扰和参考输入)

$$d_i(t) = A_d \sin(\omega_d t + \varphi_d)$$

$$\dot{d}_i = A_d \omega_d \cos(\omega_d t + \varphi_d)$$

$$\ddot{d}_i = -\omega_d{}^2 \underbrace{A_d \sin(\omega_d t + \varphi_d)}_{d_i(t)}$$

$$\Rightarrow \quad \ddot{d}_i(t) + \omega_d^2 d_i(t) = 0 \quad \text{or} \quad P_{dis}(s) = \frac{1}{s^2 + \omega_d^2} \text{ with } u = 0$$

$$s^2 D_{dis}(s) - sd_i(0) - \dot{d}_i(0) + \omega_d^2 D_{dis}(s) = 0$$

$$D_{dis}(s) = \frac{sd_i(0) + \dot{d}_i(0)}{s^2 + \omega_d^2} = P_{dis}(s)[sd_i(0) + \dot{d}_i(0)]$$

# Models of Disturbances and References

- **Mixed Step and Sinusoidal Type Disturbances and References**(混合阶跃和正弦类型的干扰和参考输入)

$$d_i(t) = A_{d0} + A_d \sin(\omega_d t + \varphi_d)$$

$$\Rightarrow \quad \dddot{d}_i(t) + \omega_d{}^2 \dot{d}_i(t) = 0 \quad \text{or} \quad P_{dis}(s) = \frac{1}{s(s^2 + \omega_d{}^2)} \text{ with } u = 0$$

- **General *Bounded* Disturbances and References** (一般有界干扰和参考输入)

$$d_i^{(q)}(t) + \gamma_{q-1} d_i^{(q-1)}(t) + \ldots + \gamma_0 d_i(t) = 0$$

$$\Rightarrow \quad P_{dis}(s) = \frac{1}{\Gamma_{dis}(s)},$$

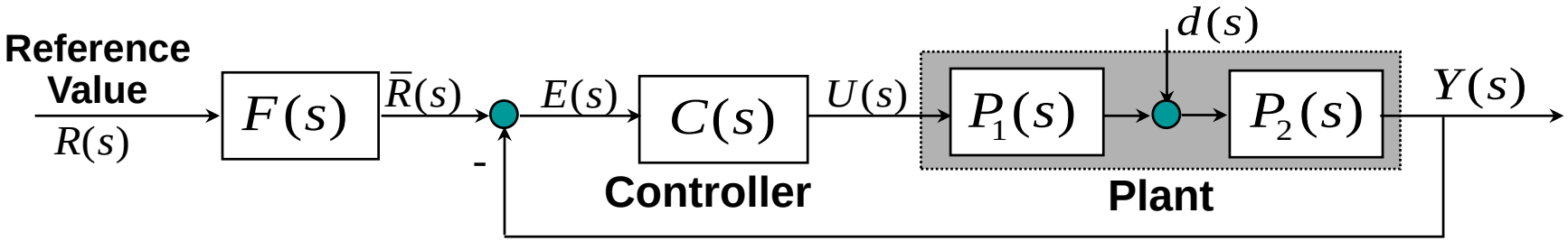
Disturbance generating polynomial (干扰生成多项式)

$$\Gamma_{dis}(s) = s^q + \gamma_{q-1} s^{q-1} + \ldots + \gamma_0$$

$$D_{dis}(s) = \frac{N_{dis}(s)}{\Gamma_{dis}(s)}$$

# Internal Model Principle for Disturbances

- **Closed-Loop System Output due to Disturbance** (因干扰引起的闭环系统输出)




$$Y(s) = S(s)P_2(s)d(s) = \frac{D_{p1}(s)\cancel{D_{p2}(s)}D_c(s)}{A_{CL}(s)}\frac{N_{p2}(s)}{\cancel{D_{p2}(s)}}\frac{N_{dis}(s)}{\Gamma_{dis}(s)}$$

$$sY(s) = s\frac{D_{p1}D_{p2}D_c}{A_{CL}(s)}\cdot\frac{N_{p2}}{D_{p2}}\cdot\frac{N_{dis}}{\Gamma_{dis}} = \frac{sD_{p1}D_c}{A_{CL}(s)}\frac{N_{p2}N_{dis}}{\Gamma_{dis}}$$

$\Rightarrow sY(s)$ has all poles (i.e., poles of $A_{CL}(s)$ or all CL poles) in LHP,

$$\Rightarrow y(t) \to y_\infty = \lim_{s\to 0} sY(s) = 0 \times \frac{D_{p1}(s)\bar{D}_c(s)N_{p2}(s)N_{dis}(s)}{A_{CL}(s)} = 0$$

- **Condition for *Zero Steady-State Output* due to** Disturbances (对干扰零稳态输出的条件)

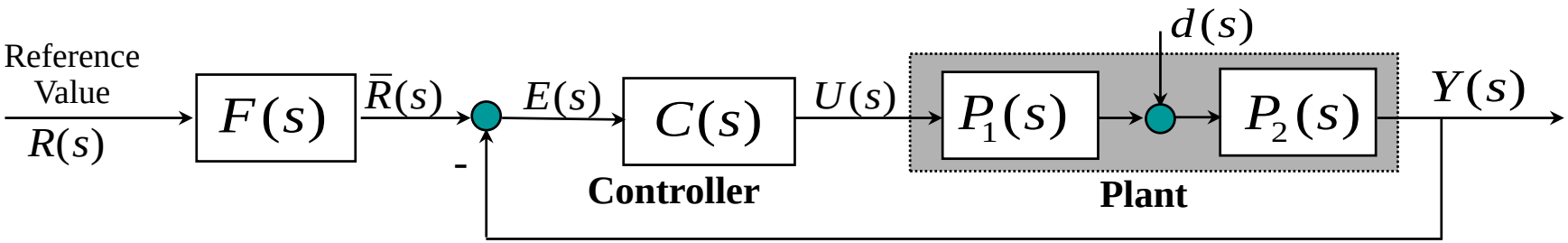
$$\boxed{D_c(s) = \Gamma_{dis}\bar{D}_c(s)}$$

# Internal Model Principle for Disturbances

- **Internal Model Principle**
  - ✓ *Steady-state disturbance compensation* requires that generating polynomial of disturbances be included as part of the controller denominator, which is called the *Internal Model Principle*. (稳态干扰补偿要求干扰生成多项式作为控制器分母的一部分，这称为内模原理)
  - ✓ Note that the roots of the generating polynomial, in particular, the ones on the imaginary axis, impose the same performance trade-offs on the closed-loop as of those poles were part of the plant ! (注意生成多项式的根特别是虚轴上的根是闭环系统极点的一部分，存在闭环性能取舍平衡问题)
- **Industrial Applications**
  - Hard Disk Drive Industry for compensation of repeatable run-outs (i.e., periodic disturbances and references due to the constant spinning speed of disks) (硬盘驱动系统的重复运动补偿(即由于磁盘的恒定旋转速度而引起的周期性干扰和参考输入))
  - Roll-eccentricity compensation in Rolling Mills, …. (轧机轧辊偏心补偿)
  - –

# Internal Model Principle for Reference Tracking

- **CL System Output Error due to References** (对参考输入引起的闭环系统输出)




$$Y(s) = T(s)F(s)R(s)$$

$$E(s) = \bar{R}(s) - Y(s) = S(s)F(s)R(s) = \frac{D_p(s)D_c(s)}{A_{CL}(s)} \frac{N_F(s)}{D_F(s)} \frac{N_r(s)}{\cancel{\Gamma_r(s)}}$$

$\cancel{\Gamma_r(s)}\bar{D}_c(s)$

- **Condition for *Zero Steady-State Error* due to References** (对参考输入零稳态误差的条件)

$$\boxed{D_c(s) = \Gamma_r(s)\bar{D}_c(s)}$$

# Internal Model Principle for RIP Control

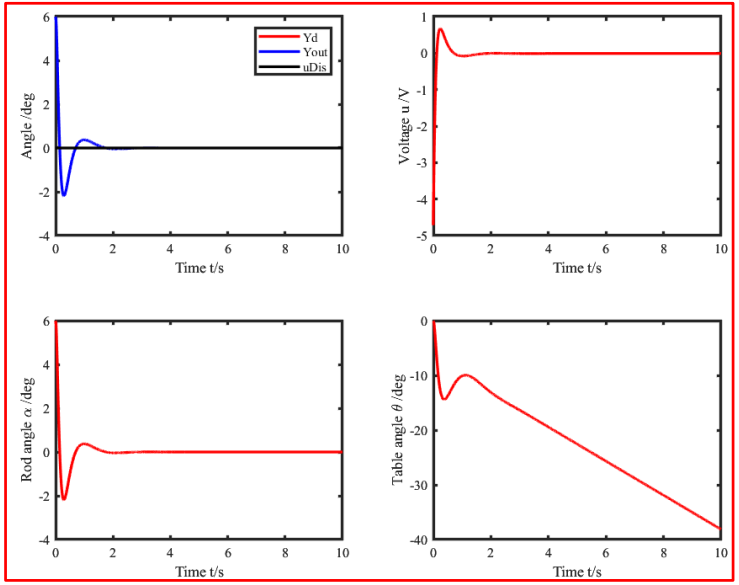
- **Balance Control of Rotary Inverted Pendulum (RIP) with Step Input Disturbance**

$$P_\alpha(s) = \frac{9.962s}{(s+12.53)(s+0.7393)(s-9.8)}$$

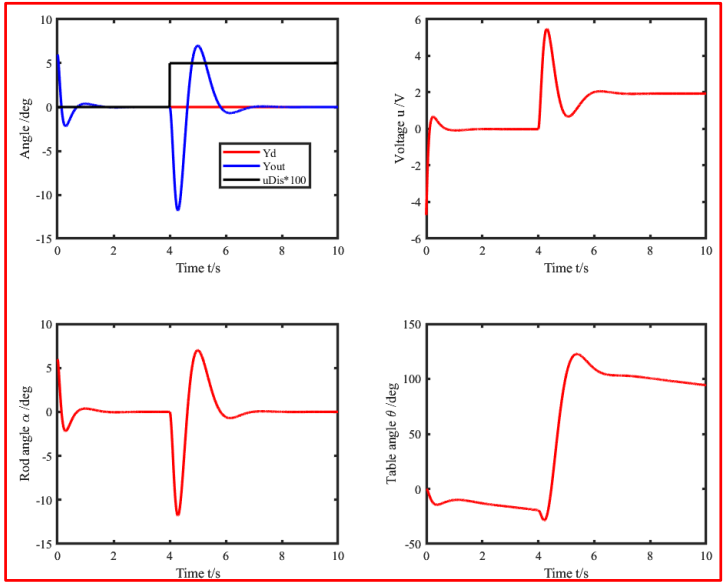
$$\Rightarrow C(s) = \frac{45.019(s+12.51)(s^2+1.673s+2.08)}{s^2(s+30.63)}$$

Performance Requirements
(i)Overshoot: OS <10%
(ii)Settling time: ts<2s
(iii)Zero steady-state error
Desired Poles of CL System
-10.0424
-10.0238 + 0.0104i
-10.0238 - 0.0104i
-2.0060 + 2.7452i
-2.0060 - 2.7452i

Although there is constant disturbance, the rod angle is return to zero as time goes!

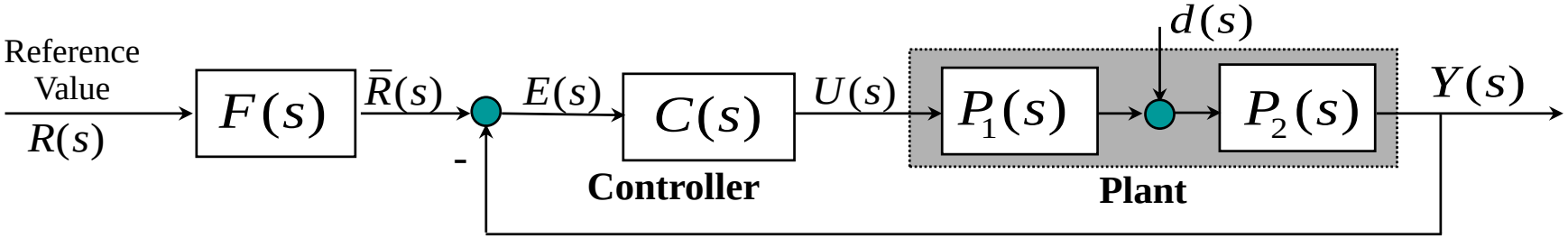



Balance control without input disturbance




Balance control with step input disturbance

# Feedforward for General Reference Tracking




- **CL System Output due to Reference Inputs** (由参考输入引起的闭环系统输出)

Ideally, perfect tracking can be achieved even during the transient if feedforward TF $F(s)$ is (理想情况下，通过选择前馈传递函数$F(s)$，即使在瞬态过程中也可以实现较好的跟踪)

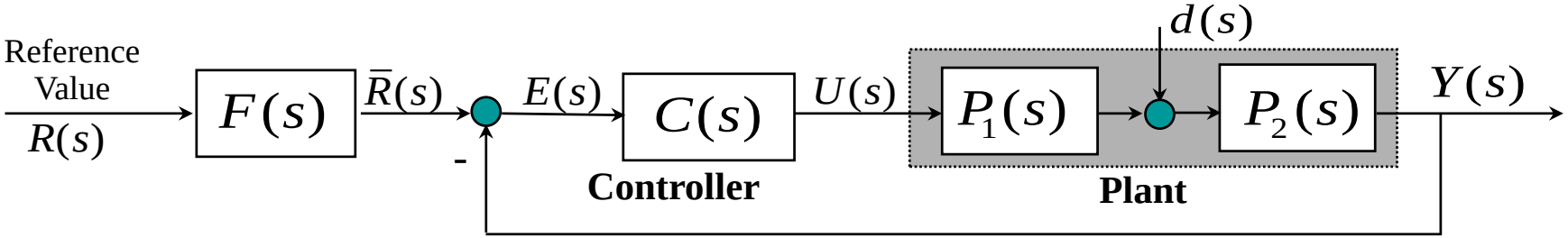
$$Y(s) = T(s)F(s)R(s) \quad \Rightarrow \quad T(s)F(s) = 1$$

In reality, the above exact inversion may not be done due to (实际情况中，可能不允许采用上述精确求逆的方法，由于):

(1) The CLTF $T(s)$ has unstable CL zeros (闭环传递函数有不稳定的闭环零点).

(2) The inversion $1/T(s)$ may not be proper and future reference input trajectory is not known, which means $F(s)=1/T(s)$ is not calculable or implementable .($T(s)$的逆可能不是正则的，并且将来的参考输入轨迹未知，这意味着 $\bar{R}(s) = R(s)/T(s)$ 无法计算或可实施)

# Feedforward for General Reference Tracking




- **CL System Output due to Reference Inputs** (由参考输入引起的闭环系统输出)

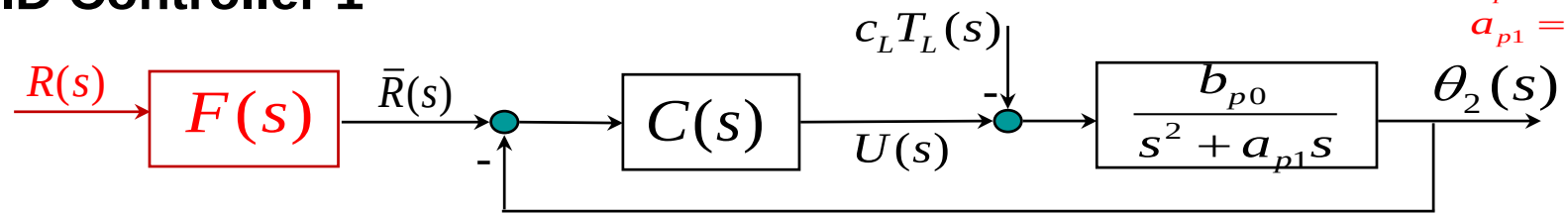
  In these cases, instead of the above exact inversion, approximated inversions may be carried out. For example, in the second case, fast poles may be inserted to make the resulting TF proper:(在这些情况下，可以用近似求逆代替上述精确求逆。例如，在第二种情况下，可增加快速极点以使得到的传递函数是正则的)

$$F(s) = \frac{1}{T(s)} \frac{1}{(\tau_h s + 1)^k}, \quad \tau_h << 1$$

# PID Control of Rigid Belt ECP Emulator

- **PID Controller 1**

$b_{p0} = 623$
$a_{p1} = 3.5$




A proper PID controller has been previously synthesized to place CL poles at **(之前设计了一个正则PID控制器，其放置闭环极点在)**

$$p_{CL1,2d} = -20 \pm 20j, \qquad p_{CL3,4d} = -40$$

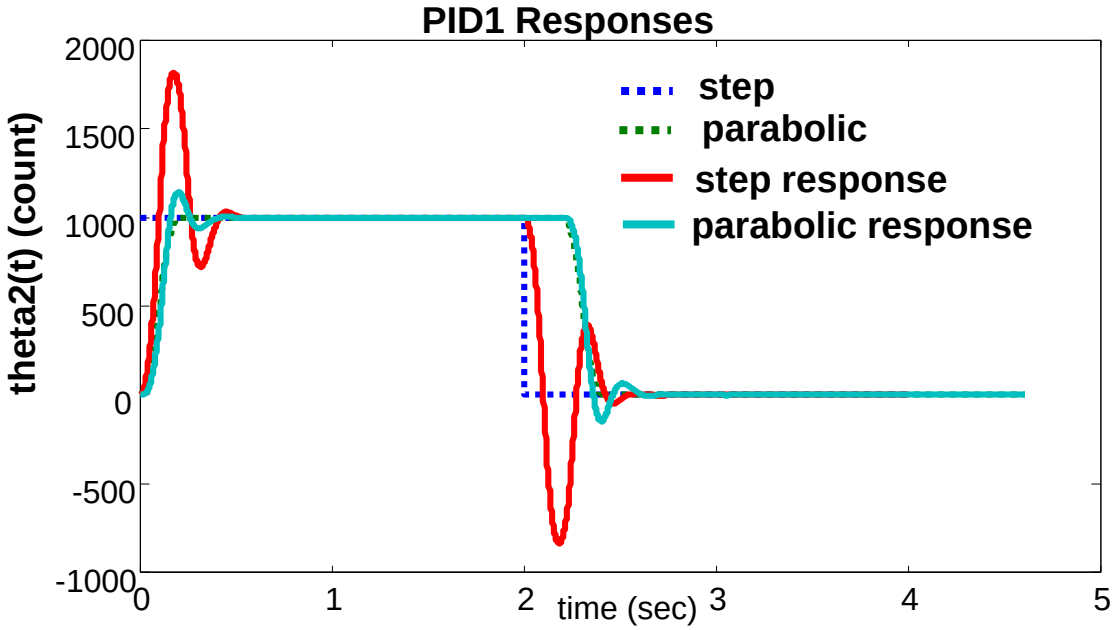
$$C(s) = \frac{8.33s^2 + 205s + 2054}{s^2 + 117s}$$

$$T_n(s) = \frac{PC}{1+PC} = \frac{N_p N_c}{A_{CL}(s)} = \frac{b_{p0}\left(8.33s^2 + 205s + 2054\right)}{\left((s+20)^2 + 20^2\right)(s+40)^2}$$

$$Y(s) = T(s)F(s)R(s)$$

# PID Control Experimental Results

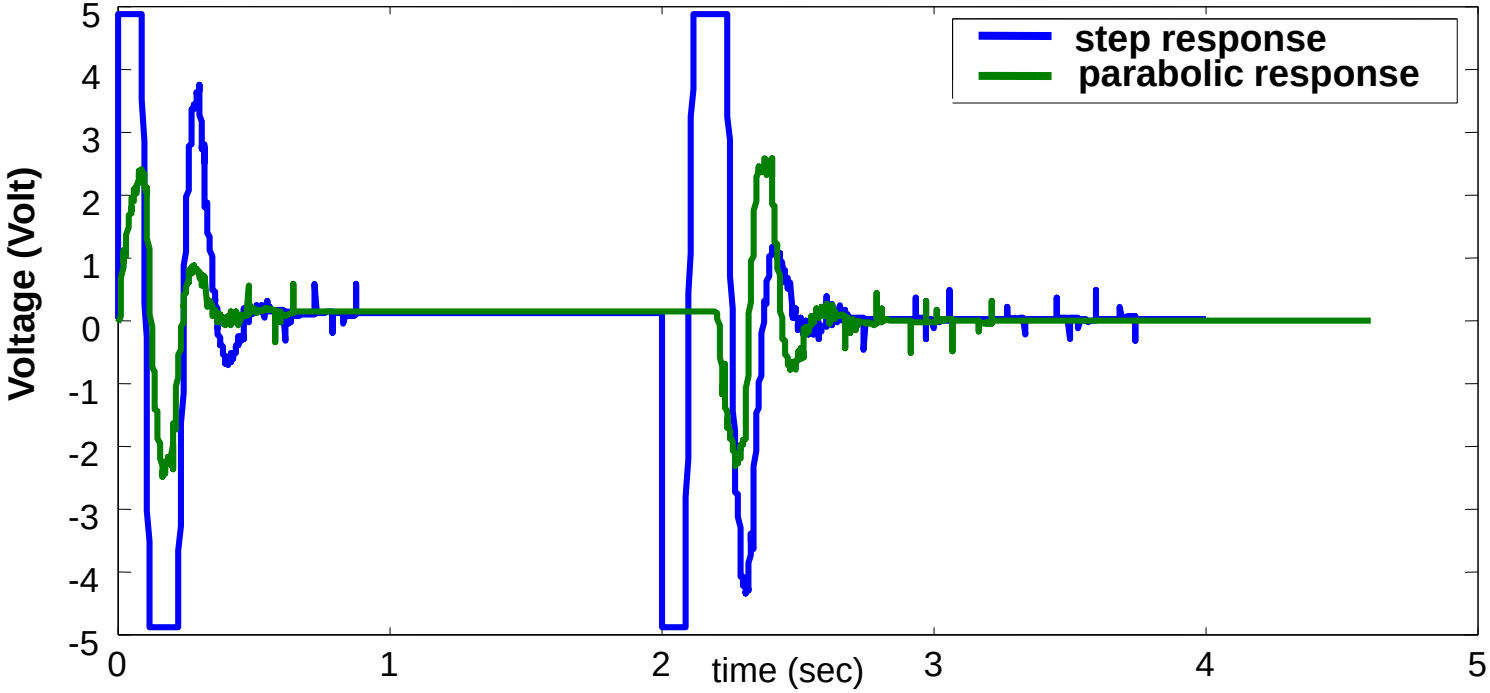
- **Trajectory Planning for Reducing/Eliminating Actuator** Saturation (减少/消除执行器饱和的轨迹规划)




The above figure shows the improved transient response of using trajectory planning (less overshoot and no degradation of settling time). (上图可看出，使用轨迹规划后瞬态响应得到了改善(超调量减少，调整时间没有降低))

# PID Control Experimental Results

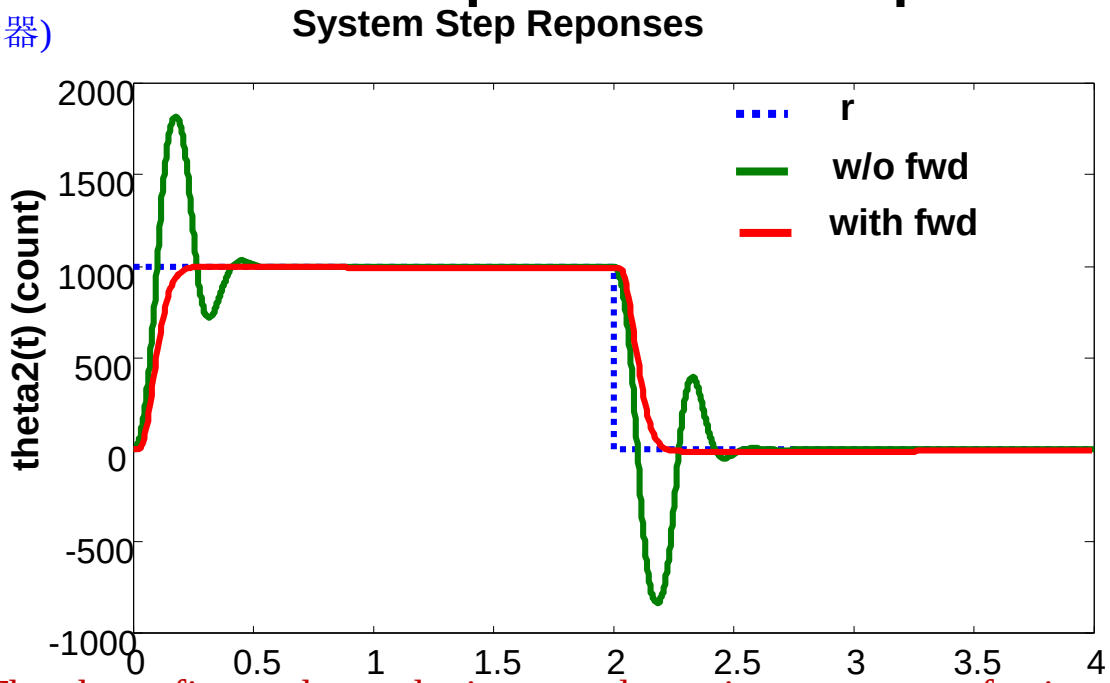
- **Control Effort** (控制量)




The above figure shows the efficacy of trajectory planning in reducing/eliminating actuator saturation (上图显示了轨迹规划在减小/消除执行器饱和方面的效果).

# PID Control Experimental Results

- **Feedforward Compensator for Improved Transient** (改进瞬态的前馈补偿器)




The comparison is done between the CL system with and without $F(s)$ in canceling the stable zeros of CLTF $T(s)$(比较了有和没有$F(s)$的闭环系统对消去闭环传函 $T(s)$稳定零点的效果)

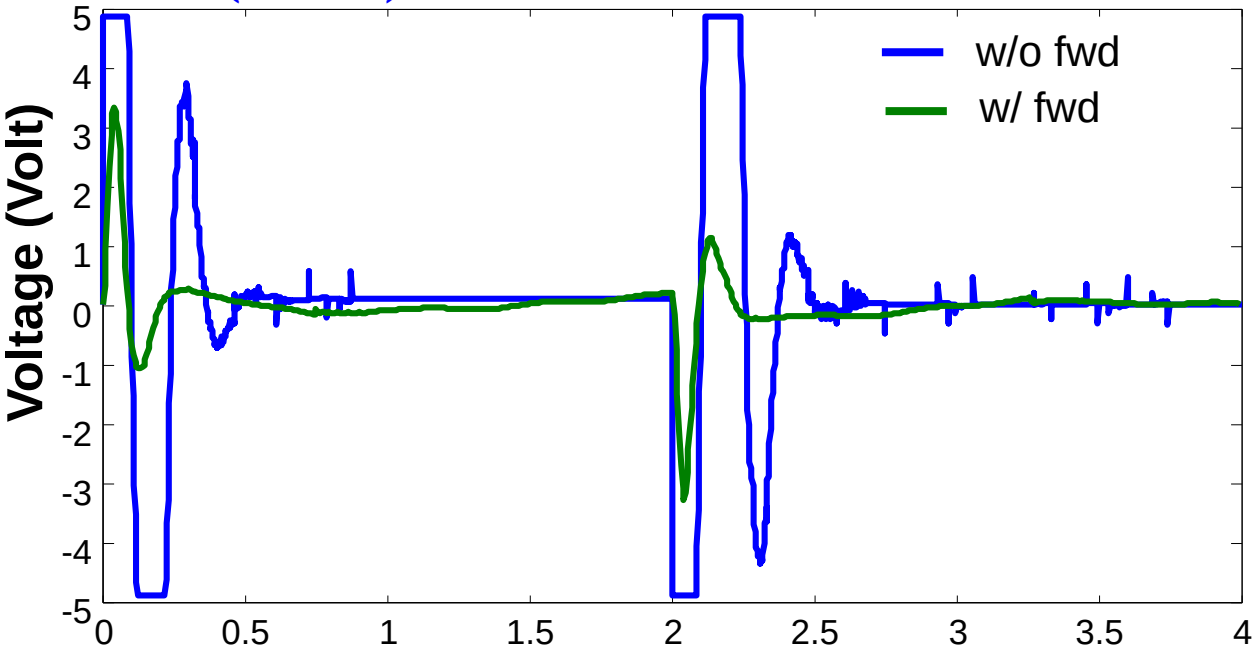
$$F(s) = \frac{246}{s^2 + 24.6s + 246}$$

$$T_n(s) = \frac{b_{p0}\left(8.33s^2 + 205s + 2054\right)}{\left((s+20)^2 + 20^2\right)(s+40)^2}$$

The above figure shows the improved transient response of using simple feedforward compensation (no overshoot vs huge overshoot) (从上图可以看出，采用简单的前馈补偿可以改善瞬态响应(无超调对比大超调)).

# PID Control Experimental Results

- **Control Effort (控制量)**

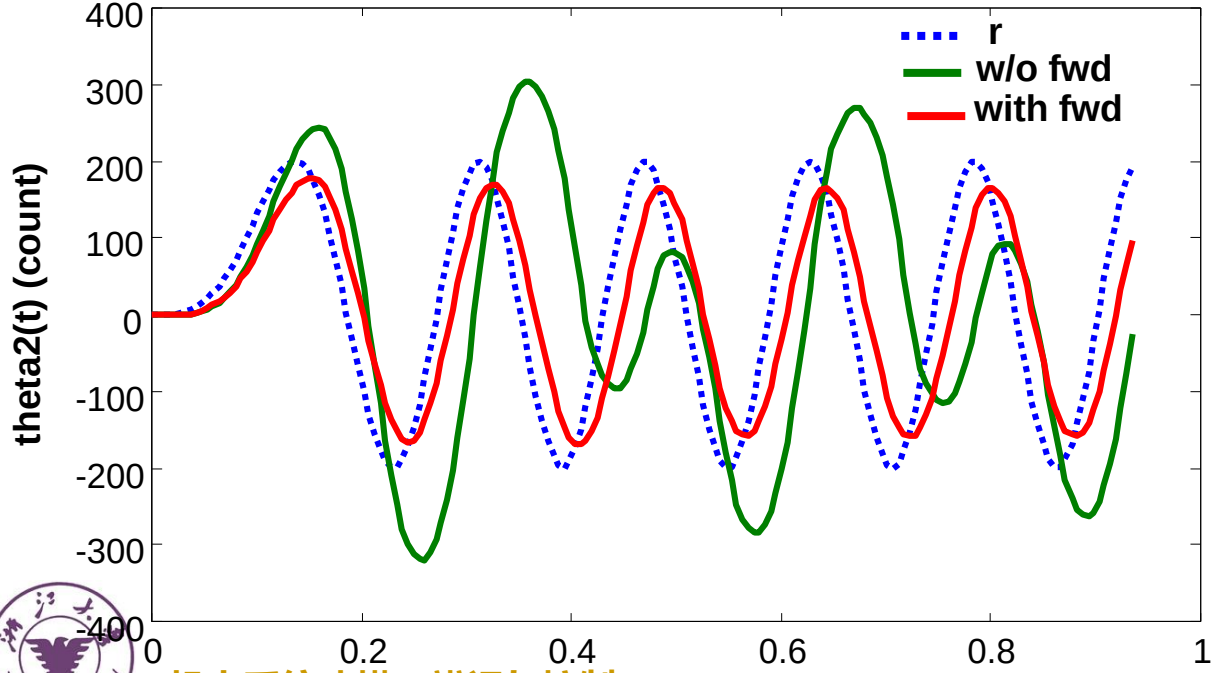



The above figure shows the efficacy of feedforward in reducing or eliminating actuator saturation due to improved transient performance (从上图可看出，由于改善了瞬态性能，前馈可以有效地减小或消除执行器饱和).

# PID Control Experimental Results

- **Feedforward Compensator for Improved Reference Tracking** (改进参考跟踪的前馈补偿器)

Tracking a sinusoidal input of $r(t)=200\sin(40t)$ counts (跟踪一个正弦参考输入)




The comparison is done between the CL system with and without $F(s)$ in canceling the stable zeros of CLTF $T(s)$ (比较了有和没有$F(s)$的闭环系统对消去闭环传函$T(s)$稳定零点的效果)

$$T_n(s)=\frac{b_{p0}\left(8.33s^2+205s+2054\right)}{\left((s+20)^2+20^2\right)(s+40)^2}$$

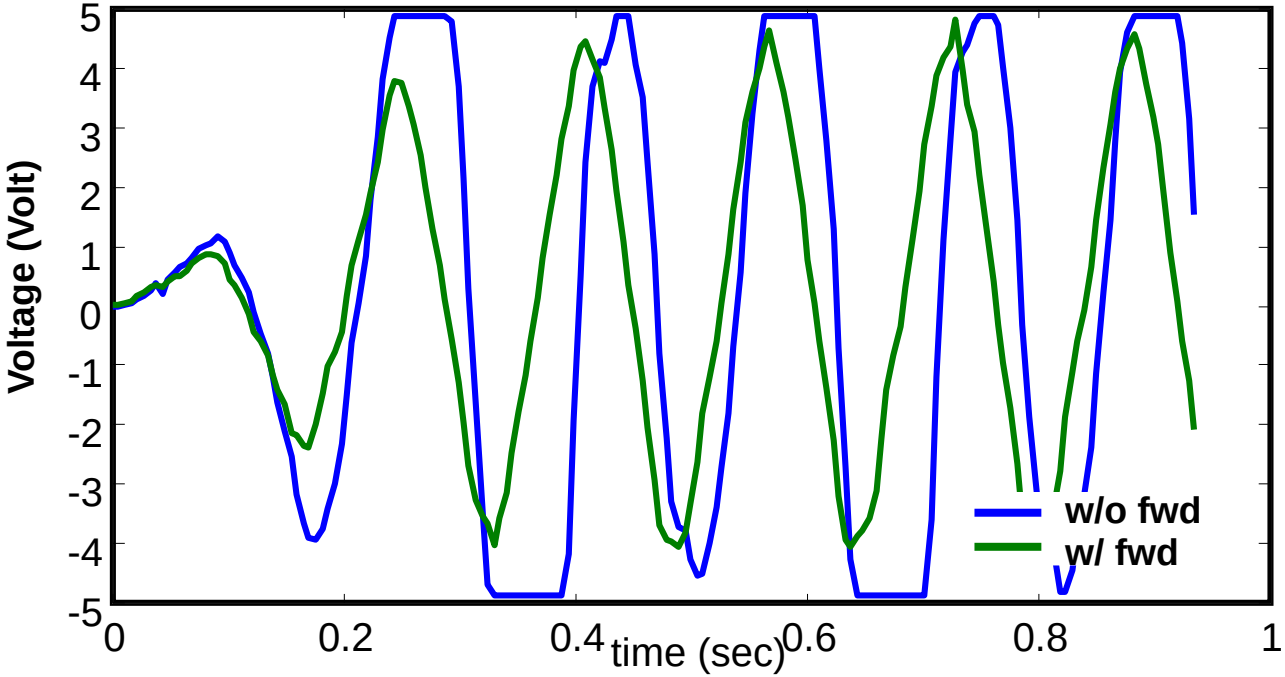
$$F(s)=\frac{1}{T(s)}\frac{1}{(\tau_h s+1)^4}$$

$$=\frac{A_d^c(s)}{8.33b_{p0}\left(s^2+24.6s+246\right)}\frac{1}{(\tau_h s+1)^4}$$

$$\tau_h=0.01$$

# PID Control Experimental Results

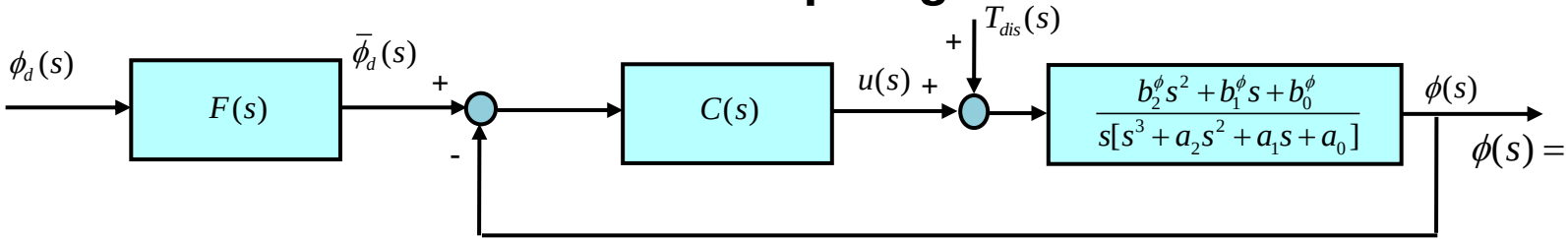
- **Control Effort** (控制量)




The above figure shows the efficacy of feedforward in reducing or eliminating actuator saturation due to improved transient performance (从上图可看出，由于改善了瞬态性能，前馈可以有效地减小或消除执行器饱和).

# Feedforward for RIP Control with Tip Angle Feedback

## ➢Balance control of RIP with Tip Angle Feedback




$$\phi(s) = 2\frac{L_c}{L_t}\alpha(s) - \theta(s)$$

$$P_{\phi}(s) = \frac{5.185s^2 - 5.172s + 452}{s(s^3 + 3.476s^2 - 120.8s \quad - 90.83s)}$$

$$= \frac{5.1848((s-0.499)^2 + 9.323^2)}{s(s+12.53)(s+0.7393)(s-9.8)}$$

$$\Rightarrow C(s) = \frac{357.9s^4 + 5403s^3 + 1.33\times10^4 s^2 + 2.49\times10^4 s + 2.6\times10^4}{s(s^3 + 60.72s^2 - 183.9s + 8835)}$$

$$= \frac{357.91(s+12.54)(s+1.625)((s-0.47)^2 + 1.83^2)}{s(s+65.67)((s-2.43)^2 + 11.35^2)}$$

CL Poles

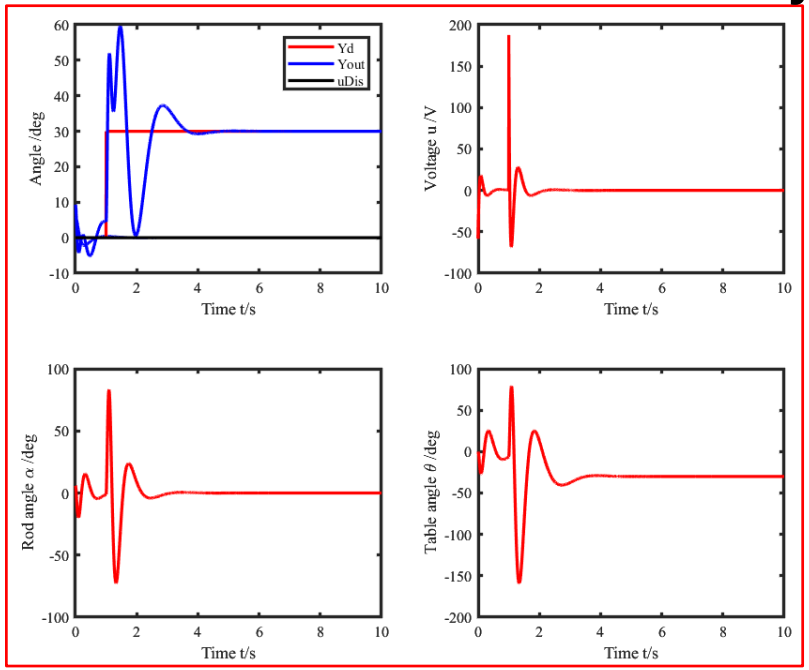
-10.9569
-9.1785
-10.4537 + 0.8014i
-10.4537 - 0.8014i
-9.5686 + 0.7363i
-9.5686 - 0.7363i
-2.0060 + 2.7452i
-2.0060 - 2.7452i

$$\Rightarrow T(s) = \frac{1855.7\boxed{(s+12.54)}\boxed{(s+1.625)}\boxed{(s^2+0.93s+3.57)}(s^2 - 0.998s + 87.17)}{(s+10.96)(s+9.18)(s^2+19.14s+92.1)(s^2+20.91s+109.9)(s^2+4.01s+11.56)}$$

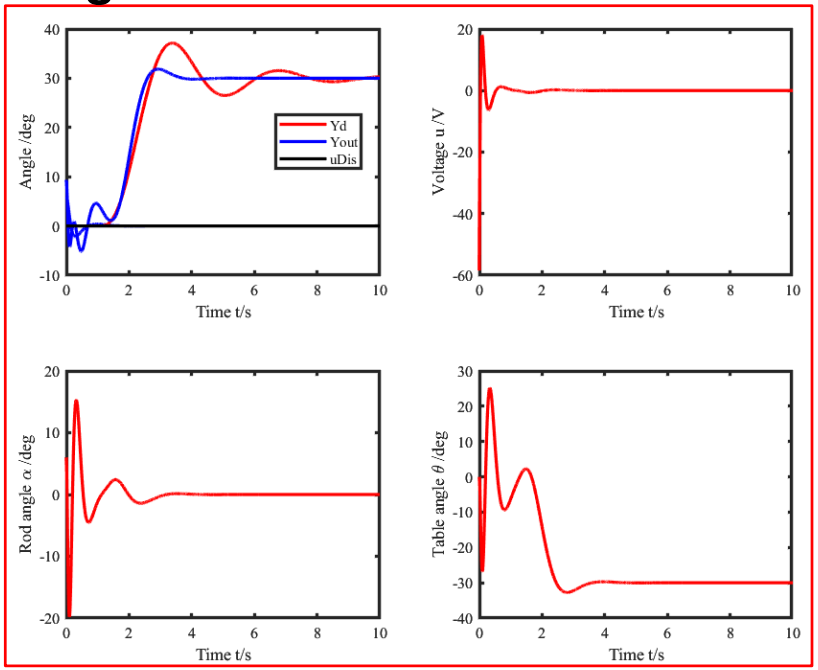
$$\Rightarrow F(s) = \frac{72.759}{(s+12.54)(s+1.625)(s^2+0.93s+3.57)}$$

# Feedforward for RIP Control with Tip Angle Feedback

## ➢Balance control of RIP with only tip angle feedback



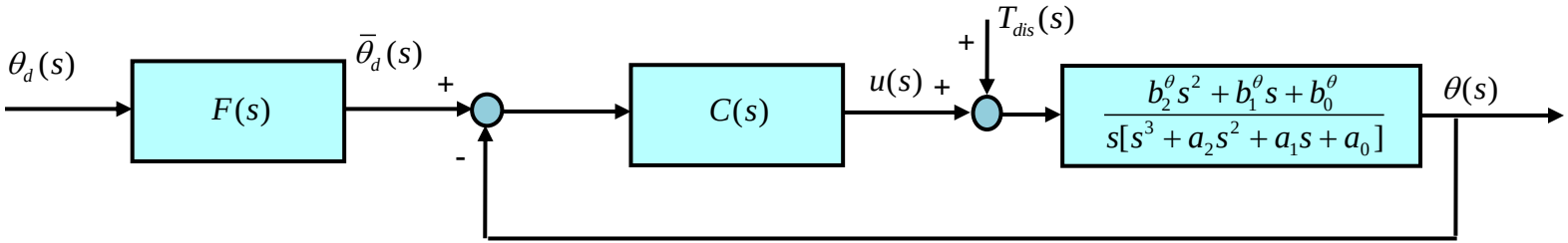

Balance control without Feedforward




Balance control with Feedforward

The transient values of table angle and voltage are reduced significantly when adding the feed forward controller due to removing the small stable open loop zeros!

# Feedforward for RIP Control with Table Angle Feedback




$$P_\phi(s) = \frac{10.36s^2 + 5.172s - 452}{s^4 + 3.476s^3 - 120.8s^2 - 90.83s}$$

$$= \frac{10.357\ (s+6.86)\ (s-6.361)}{s\ (s+12.54)\ (s-9.8)\ (s+0.7393)}$$

$$\Rightarrow C(s) = \frac{750.9s^4 + 9280s^3 - 3498s^2 - 2.49\times10^4 s - 2.60\times10^4}{s(s^3 + 60.72s^2 - 6105s - 4.441\times10^4)}$$

$$= \frac{750.86(s+12.54)(s-2.04)(s+0.93+0.70j)(s+0.93-0.70j)}{s\ (s+111.8)\ (s-57.92)\ (s+6.86)}$$

CL Poles

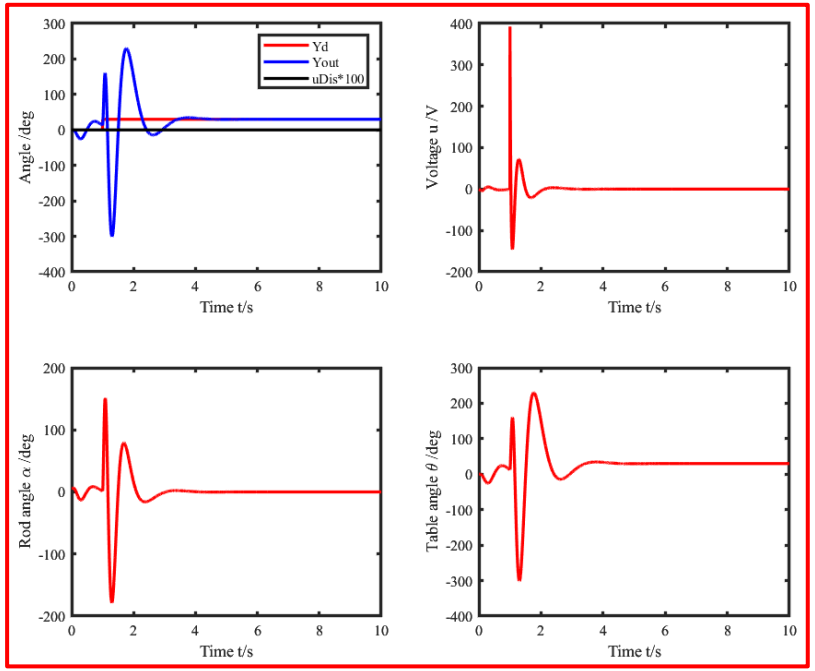
-10.9569
-9.1785
-10.4537 + 0.8014i
-10.4537 - 0.8014i
-9.5686 + 0.7363i
-9.5686 - 0.7363i
-2.0060 + 2.7452i
-2.0060 - 2.7452i

$$\Rightarrow T(s) = \frac{7776.7\boxed{(s+12.54)}\boxed{(s+6.86)}(s-6.36)(s-2.04)\boxed{(s^2+1.86s+1.36)}}{(s+10.74)(s+9.44)(s^2+19.35s+93.87)(s^2+20.65s+107)(s^2+4.01s+11.56)}$$

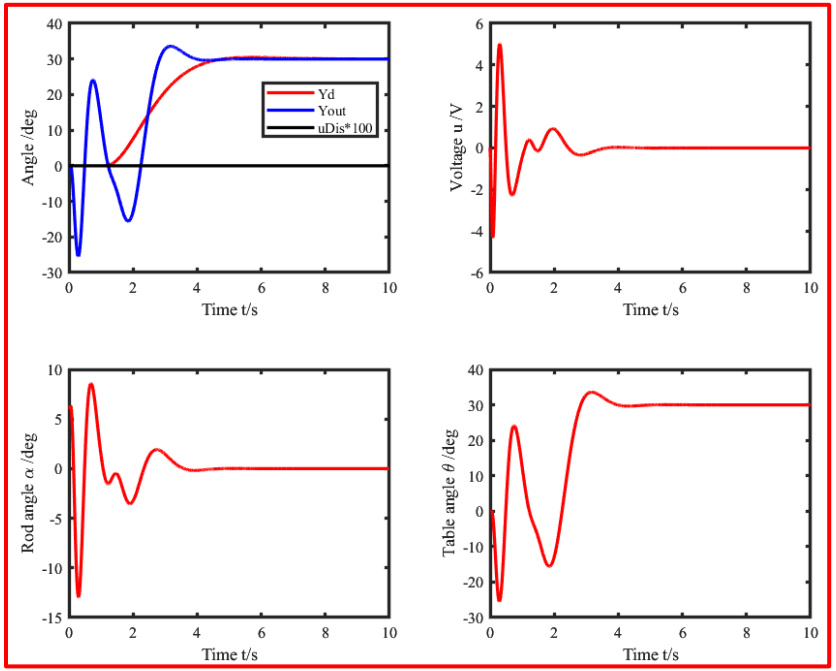
$$\Rightarrow F(s) = \frac{116.84}{(s+12.54)(s+6.86)(s^2+1.86s+1.36)}$$

# Feedforward for RIP Control with Table Angle Feedback

## ➢Balance control of RIP with only table angle feedback




Balance control without Feedforward




Balance control with Feedforward

In case of table angle feedback, the transient response of output, rod angle, voltage and table angle are greatly improved!

# Cascade Control

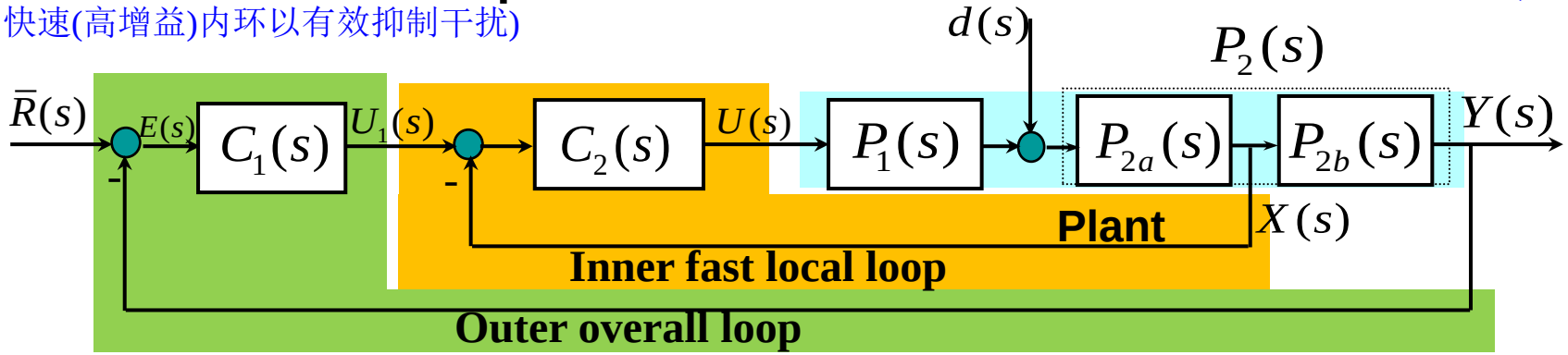
- **Motivations** (动机)
  - The previous controller design only uses output feedback and is often required to meet conflicting requirements such as the high open loop gain for good disturbance rejection capability and the low open-loop gain for good noise attenuation capability and robust stability in the presence of modeling errors, which often leads to difficult design trade-offs; **One-stone-two-birds problem is always hard to solve** (之前的控制器设计只使用输出反馈，通常需要满足相互冲突的需求,如存在建模错误时采用高开环增益以获得良好的抗干扰能力；但同时需采用低开环增益以获得良好的噪声衰减能力和存在建模误差时的鲁棒稳定性，这通常会导致较难的设计取舍问题;一石二鸟的问题总是很难解决) .

# Cascade Control

If additional feedback information (e.g., current feedback in motor driven EM systems) is available, then, one might be able to close an inner loop first to focus only on a particular local problem such as disturbance attenuation. By doing so, the overall controller design may not have to make hard trade-offs as in the traditional SISO designs (e.g., the usual motor amplifier employs local current feedback loop to make the electrical dynamics so fast that they can be neglected in the design of overall controller for drive systems like the ECP emulator). The idea is also illustrated by the following example.(如果额外的反馈信息(例如电机驱动中电磁系统的电流反馈)可用，那么可以先闭环一个内部环路以便只关注如干扰抑制等特定的局部问题。通过这样做，整个控制器的设计可能不用像传统SISO设计那样需要艰难的取舍 (例如通常电机放大器采用局部电流反馈回路，使得电动力学响应非常快，这样我们在设计伺服电机整体控制器时就可以忽略电动力学动态)。下面的示例也说明了这一思想)

# Cascade Control

- **Close a Fast Inner Loop for Effective Attenuation of Disturbances** (关闭快速(高增益)内环以有效抑制干扰)




- **Inner Fast Local Loop**(内部快速局部环路)
  - Synthesize $C_2(s)$ to achieve a high CL bandwidth from $U_1(s)$ to $X(s)$ (设计$C_2(s)$以实现从$U_1(s)$到$X(s)$的高闭环带宽)

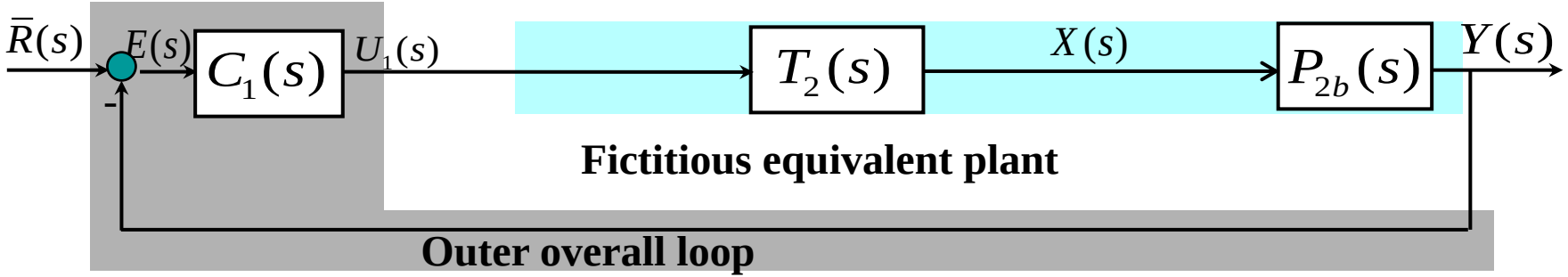
**For a large frequency range** (在较大的频率范围内)**:** $S_2(j\omega)\approx 0,\ T_2(j\omega)\approx 1$

$\Rightarrow$

$$X(s) = S_2(s)P_{2a}(s)d(s) + T_2(s)U_1(s)$$

# Two-Loop Control

- **Simplified Outer Loop Design**



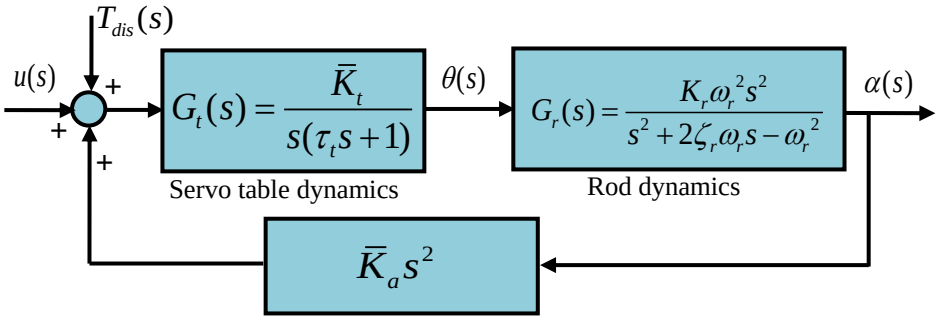

**For a large frequency range $S_2(j\omega)\approx 0$, $T_2(j\omega)\approx 1$**

# RIP Control via Both Angle Measurements

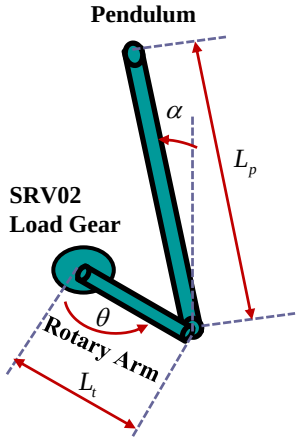
- **Problems of SISO Designs**
  - The appearance of NMP zero and its closeness to the unstable pole of the inverted pendulum dynamics significantly limits the achievable CL bandwidth, leading to poor disturbance rejection capability aside from violent transient responses.(倒立摆动力学中非最小相位零点的出现及其非常接近不稳定极点限制了可得的闭环频宽，导致系统除剧烈瞬态响应外，干扰抑制性能也较薄弱)
  - The unavoidable Coulomb frictions thus have significant effect on the CL system performance, leading to instability of the previous SISO designs in implementation.(不可避免的库伦摩擦对闭环性能有显著的影响，导致先前SISO设计在实际应用中出现不稳定现象)
- **Two-Loop Design with Both Table and Rod Angle Feedback**
  - When both table and rod angles are measured, an inner-loop design with table angle feedba ck can be conducted without the NMP zero and unstable pole issues. Thus high CL bandwidth of the inner-loop can be realized in implementation so that the Coulomb friction effect can be sufficiently attenuated to be neglected in the outer-loop design.(当同时测量悬臂和摆杆角度时，采用悬臂角度反馈的内环设计，而无需考虑NMP零点和不稳定极点问题。因此，在实施过程中可实现内环的高闭环带宽，从而充分抑制库仑摩擦影响，使其在后续外环设计中可被忽略)

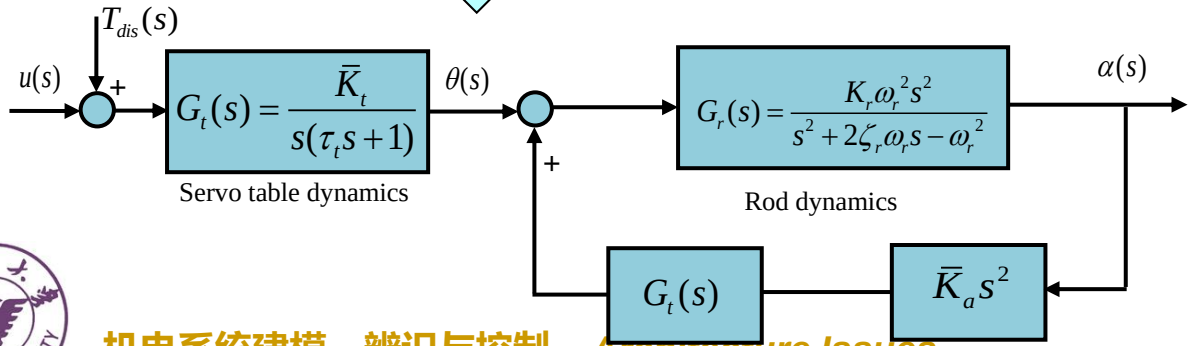
# RIP Control via Both Angle Measurements

- **System Dynamics**




$$a_3 = \tau_t - K_r K_t K_a \omega_r^2$$
$$a_2 = 2\zeta_r \omega_r \tau_t + 1$$
$$a_1 = 2\zeta_r \omega_r - \tau_t \omega_r^2$$
$$a_0 = -\omega_r^2$$



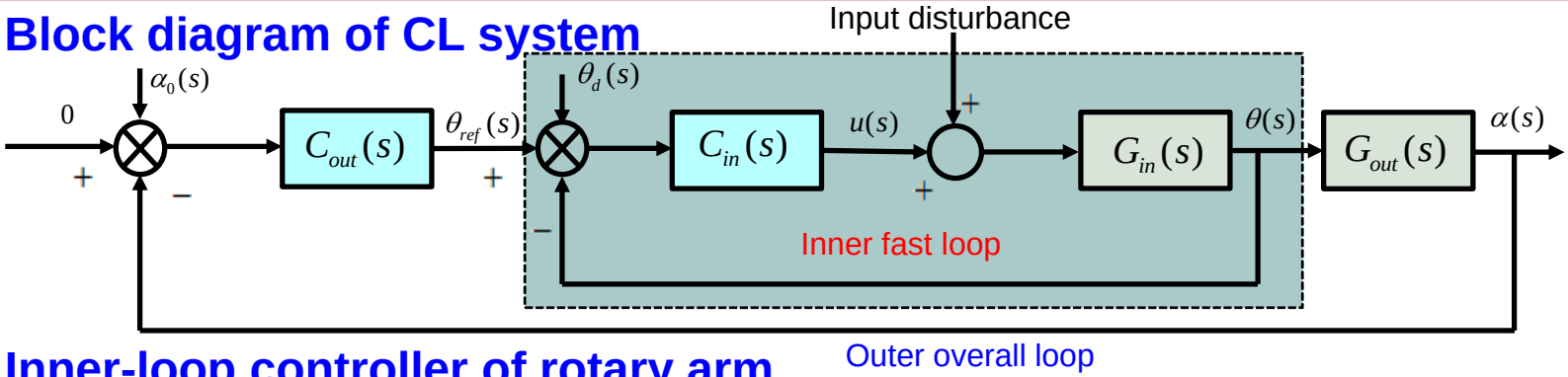

The transfer function of plant system in the inner and outer loop

$$G_{in}(s) = G_t(s) = \frac{\bar{K}_t}{s(\tau_t s+1)}$$

$$G_{out}(s) = \frac{G_r(s)}{1 - G_r(s) G_t(s) K_a s^2}$$
$$= \frac{K_r \omega_r^2 s^2 (\tau_t s+1)}{a_3 s^3 + a_2 s^2 + a_1 s + a_0}$$

# Cascade Control of Rotary Inverted Pendulum

- **Block diagram of CL system**




- **Inner-loop controller of rotary arm**

PID type controller: $C_{in}(s) = \dfrac{K_d s^2 + K_p s + K_i}{s(\tau s + 1)}$

$$G_{in}(s) = G_t(s) = \frac{\bar{K}_t}{s(\tau_t s + 1)}$$

desired bandwidth: $\omega_{n1}$=30rad/s, $\zeta_1$=0.707

desired CL pole: $p_{1,2d} = -21.21 \pm 21.21j$, $p_{3,4d} = -106.1$

- **Outer-loop controller of rod**

controller: $C_{out}(s) = \dfrac{ds^2 + es + f}{s^3 + as^2 + bs + c}$

$$G_{out}(s) = \frac{K_r \omega_r^2 s^2 (\tau_t s + 1)}{a_3 s^3 + a_2 s^2 + a_1 s + a_0}$$

desired bandwidth: $\omega_{n2}$=3rad/s, $\zeta_2$=0.707

desired CL pole: $p_{1,2d} = -2.12 \pm 2.12j$, $p_{3-6d} = -10.6$

# RIP Control via Both Angle Measurements

- **Inner Loop Design**

$$G_t(s) = \frac{K_t}{s(\tau_t s + 1)} \Rightarrow C_1(s) = \frac{K_d s^2 + K_p s + K_i}{s(\tau s + 1)} = \frac{b_{c2} s^2 + b_{c1} s + b_{c0}}{s(s + a_{c0})}$$

PID+ filter  Generic 2nd controller with integrator（带积分的2阶控制器）

$$G_t(s) = \frac{4.976}{s(1.342s + 1)} \Rightarrow C_{in}(s) = \frac{5650.9(s^2 + 31.88s + 483)}{s(s + 253.8)} \qquad F_{in}(s) = \frac{483.04}{(s^2 + 31.88s + 483)}$$

$$p_{1,2d} = -21.21 \pm j21.21$$

$$p_{3,4d} = -106.05$$

- Should be placed to make the CL fast enough that the Coulomb friction effect and other input disturbance are attenuated sufficiently（使闭环较快以便足够抑制库伦摩擦和其他输入干扰）
- High local CL bandwidth is achievable in practice as there are no NMP zeros and unstable poles（由于内环没有非最小相位零点和不稳定极点，在实际中可获得高局部闭环带宽）

# RIP Control via Both Angle Measurements

- **Outer Loop Design**

The outer loop design is to stabilize the rod movement to have a stable overall CL system（外环设计用来稳定摆杆运动以便实现稳定的闭环系统）

$$G_{out}(s) = \frac{2.6863s^2(s+0.7453)}{(s+12.54)\,(s-9.8)\,(s+0.7393)}$$

Generic 3nd controller

$$C_{out}(s) = \frac{ds^2+es+f}{s^3+as^2+bs+c}$$

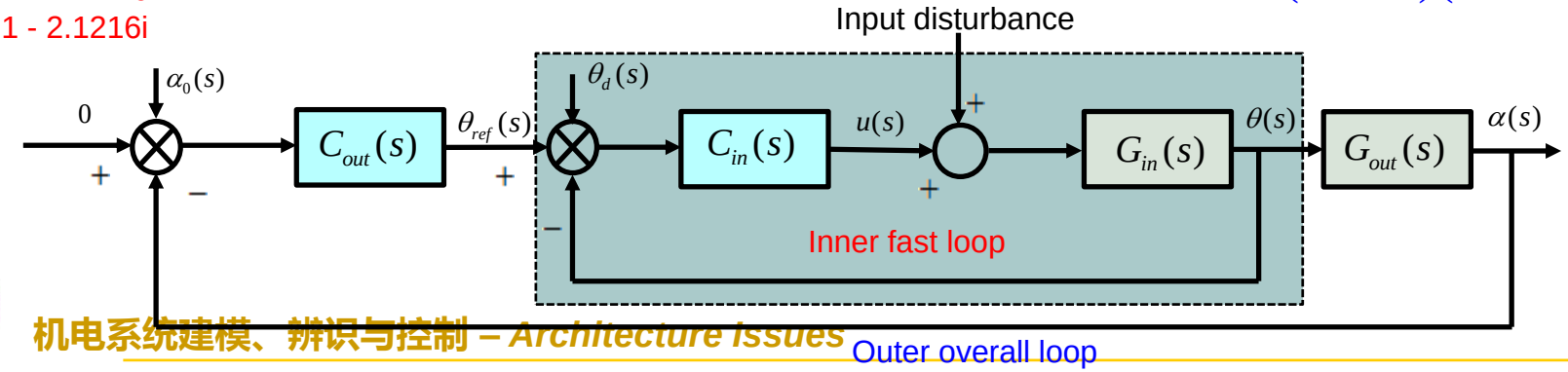
CL Poles
-10.72
-10.629 + 0.14329i
-10.629 - 0.14329i
-10.443
-2.121 + 2.1216i
-2.121 - 2.1216i

$$C_{out}(s) = \frac{-5.57\times10^4 s^2 - 1.52\times10^5 s + 6.85\times10^6}{s^3+1.5\times10^5 s^2+603.9s-1253}$$

$$= \frac{-55694\,(s+12.54)\,(s-9.808)}{(s+1.5\times10^5)\,(s+0.09355)\,(s-0.08952)}$$

$$F_{out}(s) = \frac{-9.343}{s^2+13.28s+9.343}$$

$$= \frac{-9.343}{(s+12.54)\,(s+0.7453)}$$



# Cascade Control of RIP

- **Two loop controller design**
  - **Fast local inner loop design for good friction rejection**
    - **Inner OL plant:** $P_2(s) = \dfrac{4.976}{s(1.342s+1)}$
    - **Inner controller:** $C_2(s) = \dfrac{b_{c2}s^2 + b_{c1}s + b_{c0}}{s(s+a_{c0})} = \dfrac{5650.9(s^2+31.88s+483)}{s(s+253.8)}$
  - **Outer loop design for good reference tracking**
    - **outer plant:** $P_1(s) = \dfrac{2.6863s^2(s+0.7453)}{(s+12.54)\ (s-9.8)\ (s+0.7393)}$
    - **outer controller:** $C_1(s) = \dfrac{-55694\ (s+12.54)\ (s-9.808)}{(s+1.5\times10^5)\ (s+0.09355)\ (s-0.08952)}$
  - **Feedforward design for improving transient performance**
    - **CL plant:** $CLTF\quad T = \dfrac{-1.4961\times10^5 s^2(s+12.54)(s-9.808)(s+0.7453)}{(s+10.72)(s+10.44)[(s+10.63)^2+0.143^2)+(s+2.12)^2+2.12^2)}$

$$F_1(s) = \frac{-9.3434}{(s+12.54))(s+0.7453)}$$

- **CL poles of Inner-loop control (C2(s)):**

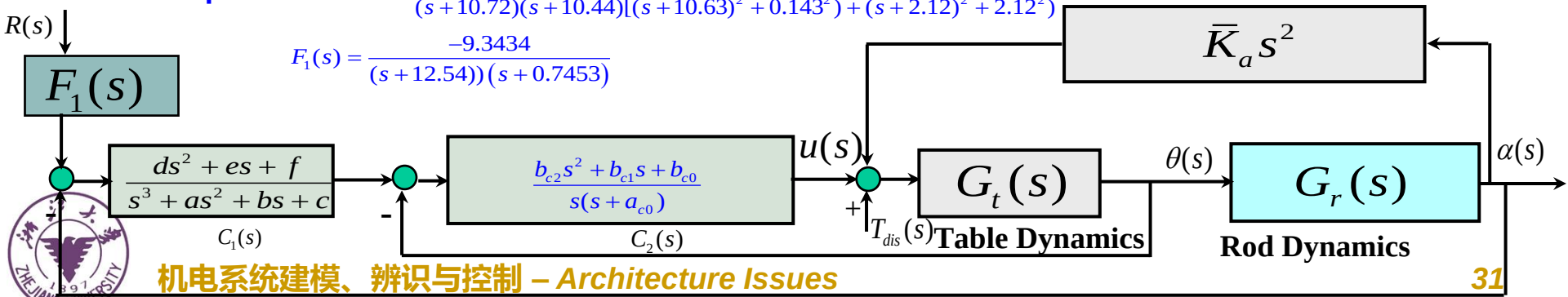
$$p_{1,2d} = -21.21 \pm j21.21$$

$$p_{3,4d} = -106.05$$

- **CL poles of Outer-loop control (C1(s)):**

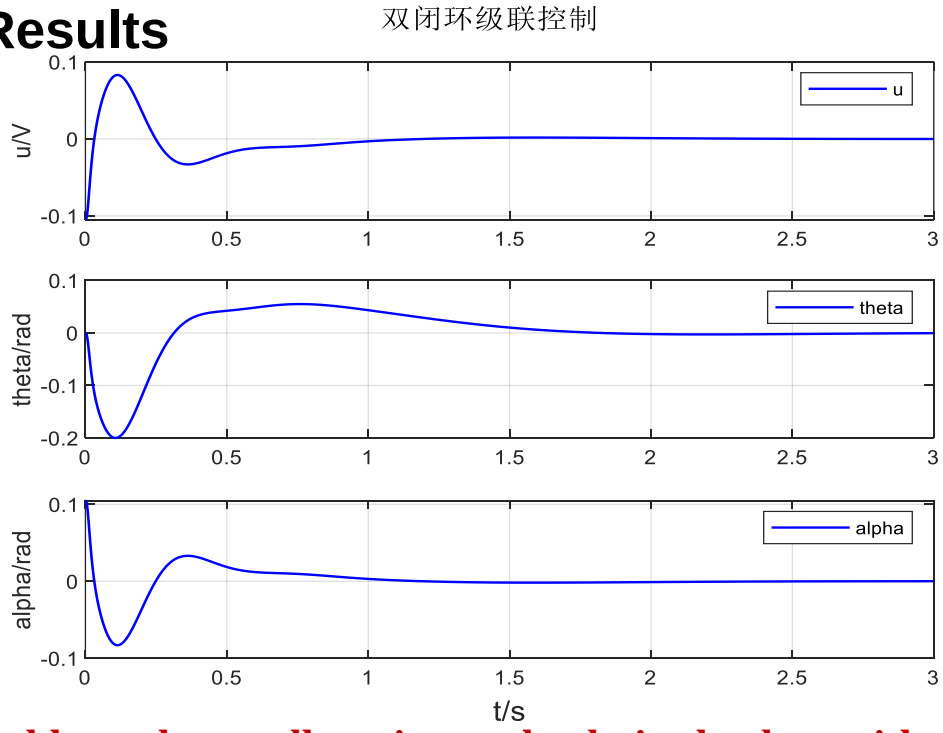
$$p_{CLd1,2} = -2.12 \pm 2.12\,j,$$

$$p_{CLd3-6} = -10.63$$



# Cascade Control of RIP

- **Simulation Results**




**Rod angle and table angle are all getting to the desired values with appropriate control efforts.**